

Project Task 2: Processor Design Semester Project

Project Task 2-Folder contents

- Project2_ekene.py:
- README.md
- Project output file (This file)

How to run processor:

1. Download project from ICollege or GitHub
2. CD to the project folder that contains: project 2 folder
3. Run the following command: **python project2_ekene.py**
4. Enter the number of variables: (must be greater or equal to 2)
5. Enter the numbers for each row with exactly one space between the number
6. When done with a row, press enter to input numbers to the next row.
7. When done you should see a question asking if you want in SOP or POS form
8. Enter the SOP for SOP form or POS for POS form then click enter in terminal
9. You should be able to see the canonical form, the minterms/maxterms, K-Map, the simplified expression, and whether it is a pass or fail.

Scenario one

Input variable is 2

Row	Inputs (A B)	Output
Row 1	00	0
Row 2	01	1
Row 3	10	1
Row 4	11	1

SOP Output:

```

Enter number of variables (>=2): 2
Enter truth table rows (inputs then output)
Example for 3 variables would be typed like this: 0 1 1 0
Row 1: 0 0 0
Row 2: 0 1 1
Row 3: 1 0 1
Row 4: 1 1 1

Truth Table
(0, 0) -> 0
(0, 1) -> 1
(1, 0) -> 1
(1, 1) -> 1

Choose form (SOP or POS): sop

Canonical SOP:
A'B + AB' + AB
Minterms: [1, 2, 3]

K-Map
      B
      0 1
A 0   0 1
A 1   1 1

Simplified Expression:
B + A
Row 1: Input (0, 0) : Output 0 [MATCH]
Row 2: Input (0, 1) : Output 1 [MATCH]
Row 3: Input (1, 0) : Output 1 [MATCH]
Row 4: Input (1, 1) : Output 1 [MATCH]

Validation Result: PASS
The simplified expression is equal to the truth table.
PS C:\Users\user\OneDrive\Desktop\fall 2025\Spring 2026\Computer Architecture\Computer-Architecture-Project\project 2>

```

POS Output:

```

PS C:\Users\user\OneDrive\Desktop\fall 2025\Spring 2026\Computer Architecture\Computer-Architecture-Project\project 2> python e.py
Enter number of variables (>=2): 2
Enter truth table rows (inputs then output)
Example for 3 variables would be typed like this: 0 1 1 0
Row 1: 0 0 0
Row 2: 0 1 1
Row 3: 1 0 1
Row 4: 1 1 1

Truth Table
(0, 0) -> 0
(0, 1) -> 1
(1, 0) -> 1
(1, 1) -> 1

Choose form (SOP or POS): pos

Canonical POS:
(A + B)
Maxterms: [0]

K-Map
      B
      0 1
A 0   0 1
A 1   1 1

Simplified Expression:
(A + B)
Row 1: Input (0, 0) : Output 0 [MATCH]
Row 2: Input (0, 1) : Output 1 [MATCH]
Row 3: Input (1, 0) : Output 1 [MATCH]
Row 4: Input (1, 1) : Output 1 [MATCH]

Validation Result: PASS
The simplified expression is equal to the truth table.
PS C:\Users\user\OneDrive\Desktop\fall 2025\Spring 2026\Computer Architecture\Computer-Architecture-Project\project 2>

```

Scenario 2

Input variable is 3

Row	Inputs (A B C)	Output
Row 1	000	0
Row 2	001	0
Row 3	010	0
Row 4	011	1
Row 5	100	1
Row 6	101	1
Row 7	110	1
Row 8	111	1

SOP output:

```

Enter number of variables (>=2): 3
Enter truth table rows (inputs then output)
Example for 3 variables would be typed like this: 0 1 1 0
Row 1: 0 0 0 0
Row 2: 0 0 1 0
Row 3: 0 1 0 0
Row 4: 0 1 1 1
Row 5: 1 0 0 1
Row 6: 1 0 1 1
Row 7: 1 1 0 1
Row 8: 1 1 1 1

Truth Table
(0, 0, 0) -> 0
(0, 0, 1) -> 0
(0, 1, 0) -> 0
(0, 1, 1) -> 1
(1, 0, 0) -> 1
(1, 0, 1) -> 1
(1, 1, 0) -> 1
(1, 1, 1) -> 1

Choose form (SOP or POS): sop

Canonical SOP:
A'BC + AB'C' + AB'C + ABC' + ABC
Minterms: [3, 4, 5, 6, 7]

K-Map
      BC
      00 01 11 10
A=0   0 0 1 0
A=1   1 1 1 1

Simplified Expression:
BC + A
Row 1: Input (0, 0, 0) : Output 0 [MATCH]
Row 2: Input (0, 0, 1) : Output 0 [MATCH]
Row 3: Input (0, 1, 0) : Output 0 [MATCH]
Row 4: Input (0, 1, 1) : Output 1 [MATCH]
Row 5: Input (1, 0, 0) : Output 1 [MATCH]
Row 6: Input (1, 0, 1) : Output 1 [MATCH]
Row 7: Input (1, 1, 0) : Output 1 [MATCH]
Row 8: Input (1, 1, 1) : Output 1 [MATCH]

Validation Result: PASS
The simplified expression is equal to the truth table.
PS C:\Users\user\OneDrive\Desktop\fall 2025\Spring 2026\Computer Architecture\

```

POS output:

```

\Computer-Architecture-Project\project 2> python project2_ekene.py
Enter number of variables (>=2): 3
Enter truth table rows (inputs then output)
Example for 3 variables would be typed like this: 0 1 1 0
Row 1: 0 0 0 0
Row 2: 0 0 1 0
Row 3: 0 1 0 0
Row 4: 0 1 1 1
Row 5: 1 0 0 1
Row 6: 1 0 1 1
Row 7: 1 1 0 1
Row 8: 1 1 1 1

Truth Table
(0, 0, 0) -> 0
(0, 0, 1) -> 0
(0, 1, 0) -> 0
(0, 1, 1) -> 1
(1, 0, 0) -> 1
(1, 0, 1) -> 1
(1, 1, 0) -> 1
(1, 1, 1) -> 1

Choose form (SOP or POS): pos

Canonical POS:
(A + B + C)(A + B + C')(A + B' + C)
Maxterms: [0, 1, 2]

K-Map
      BC
      00 01 11 10
A=0   0 0 1 0
A=1   1 1 1 1

Simplified Expression:
(A + C)(A + B)
Row 1: Input (0, 0, 0) : Output 0 [MATCH]
Row 2: Input (0, 0, 1) : Output 0 [MATCH]
Row 3: Input (0, 1, 0) : Output 0 [MATCH]
Row 4: Input (0, 1, 1) : Output 1 [MATCH]
Row 5: Input (1, 0, 0) : Output 1 [MATCH]
Row 6: Input (1, 0, 1) : Output 1 [MATCH]
Row 7: Input (1, 1, 0) : Output 1 [MATCH]
Row 8: Input (1, 1, 1) : Output 1 [MATCH]

Validation Result: PASS
The simplified expression is equal to the truth table.
PS C:\Users\user\OneDrive\Desktop\fall 2025\Spring 2026\Computer Archite

```

Scenario 3

Input variable is 3

Row	Inputs (A B C D)	Output
Row 1 and Row 2	0000, 0001	1
Row 3 and Row 4	0010, 0011	0
Row 5 and Row 6	0100, 0101	1
Row 7 and Row 8	0110, 0111	0

Row 9 and Row 10	1000, 1001	1
Row 11 and Row 12	1010, 1011	0
Row 13 and Row 14	1100, 1101	1
Row 15 and Row 16	1110, 1111	0

SOP output:

```

\Computer-Architecture-Project\project 2> python project2_ekene.py
Enter number of variables (>=2): 4
Enter truth table rows (inputs then output)
Example for 3 variables would be typed like this: 0 1 1 0
Row 1: 0 0 0 0 1
Row 2: 0 0 0 1 1
Row 3: 0 0 1 0 0
Row 4: 0 0 1 1 0
Row 5: 0 1 0 0 1
Row 6: 0 1 0 1 1
Row 7: 0 1 1 0 0
Row 8: 0 1 1 1 0
Row 9: 1 0 0 0 1
Row 10: 1 0 0 1 1
Row 11: 1 0 1 0 0
Row 12: 1 0 1 1 0
Row 13: 1 1 0 0 1
Row 14: 1 1 0 1 1
Row 15: 1 1 1 0 0
Row 16: 1 1 1 1 0

Truth Table
(0, 0, 0, 0) -> 1
(0, 0, 0, 1) -> 1
(0, 0, 1, 0) -> 0
(0, 0, 1, 1) -> 0
(0, 1, 0, 0) -> 1
(0, 1, 0, 1) -> 1
(0, 1, 1, 0) -> 0
(0, 1, 1, 1) -> 0
(1, 0, 0, 0) -> 1
(1, 0, 0, 1) -> 1
(1, 0, 1, 0) -> 0
(1, 0, 1, 1) -> 0
(1, 1, 0, 0) -> 1
(1, 1, 0, 1) -> 1
(1, 1, 1, 0) -> 0
(1, 1, 1, 1) -> 0

Choose form (SOP or POS): sop

Canonical SOP:
A'B'C'D' + A'B'C'D + A'BC'D' + A'BC'D + AB'C'D' + AB'C'D + ABC'D' + ABC'D
Minterms: [0, 1, 4, 5, 8, 9, 12, 13]

K-Map
      CD
      00 01 11 10
-----
AB=00 | 1 1 0 0
AB=01 | 1 1 0 0
AB=11 | 1 1 0 0
AB=10 | 1 1 0 0

Simplified Expression:
C'

```

POS output

```

PS C:\Users\user\OneDrive\Desktop\Fall 2025\Spring 2026\Computer Architecture\Computer Architecture-Project\project 2> python project2_ekene.py
Enter number of variables (>=2): 4
Enter truth table rows (inputs then output)
Example for 3 variables would be typed like this: 0 1 1 0
Row 1: 0 0 0 0 1
Row 2: 0 0 0 1 1
Row 3: 0 0 1 0 0
Row 4: 0 0 1 1 0
Row 5: 0 1 0 0 1
Row 6: 0 1 0 1 1
Row 7: 0 1 1 0 0
Row 8: 0 1 1 1 0
Row 9: 1 0 0 0 1
Row 10: 1 0 0 1 1
Row 11: 1 0 1 0 0
Row 12: 1 0 1 1 0
Row 13: 1 1 0 0 1
Row 14: 1 1 0 1 1
Row 15: 1 1 1 0 0
Row 16: 1 1 1 1 0

Truth Table
(0, 0, 0, 0) -> 1
(0, 0, 0, 1) -> 1
(0, 0, 1, 0) -> 0
(0, 0, 1, 1) -> 0
(0, 1, 0, 0) -> 1
(0, 1, 0, 1) -> 1
(0, 1, 1, 0) -> 0
(0, 1, 1, 1) -> 0
(1, 0, 0, 0) -> 1
(1, 0, 0, 1) -> 1
(1, 0, 1, 0) -> 0
(1, 0, 1, 1) -> 0
(1, 1, 0, 0) -> 1
(1, 1, 0, 1) -> 1
(1, 1, 1, 0) -> 0
(1, 1, 1, 1) -> 0

Choose form (SOP or POS): pos

Canonical POS:
(A + B + C' + D)(A + B + C' + D')(A + B' + C' + D)(A + B' + C' + D')(A' + B + C' + D + B + C' + D')(A' + B' + C' + D)(A' + B' + C' + D')
Maxterms: [2, 3, 6, 7, 10, 11, 14, 15]

K-Map
      CD
      00 01 11 10
      -----
AB=00 | 1 1 0 0
AB=01 | 1 1 0 0
AB=11 | 1 1 0 0
AB=10 | 1 1 0 0

Simplified Expression:
(C')
Row 1: Input (0, 0, 0, 0) : Output 1 [MATCH]
Row 2: Input (0, 0, 0, 1) : Output 1 [MATCH]
Row 3: Input (0, 0, 1, 0) : Output 0 [MATCH]
Row 4: Input (0, 0, 1, 1) : Output 0 [MATCH]
Row 5: Input (0, 1, 0, 0) : Output 1 [MATCH]
Row 6: Input (0, 1, 0, 1) : Output 1 [MATCH]
Row 7: Input (0, 1, 1, 0) : Output 0 [MATCH]
Row 8: Input (0, 1, 1, 1) : Output 0 [MATCH]
Row 9: Input (1, 0, 0, 0) : Output 1 [MATCH]
Row 10: Input (1, 0, 0, 1) : Output 1 [MATCH]
Row 11: Input (1, 0, 1, 0) : Output 0 [MATCH]
Row 12: Input (1, 0, 1, 1) : Output 0 [MATCH]

```